

Mello : A Calm Companion for Stress Relief - A Multi-sensory and Interactive Plush Toy

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Abstract

Mello is a multi-sensory interactive plush toy designed to support emotional well-being and alleviate stress through a combination of tactile, haptic, auditory, and visual stimuli. Mello integrates a pressure-sensitive interface, vibrotactile feedback, ambient soundscapes, and a pulsing LED light to offer a calming user experience. An accompanying mobile application allows for real-time control and customization of sounds, tracking of usage behavior, and personal adjustment of features. A user-centered design process involving iterative prototyping and user feedback shaped Mello's development. Evaluation through user testing (n=15) showed high satisfaction (4.2/5), with participants reporting feelings of joy, calm, or groundedness. Mello demonstrates the potential of tangible interfaces in emotional regulation and everyday stress relief; and opens avenues for future work in multi-sensory & multi-modal affective interaction design.

CCS Concepts

•Human-centered computing ~Human computer interaction (HCI) ~Interaction devices ~Haptic devices •Human-centered computing ~Human computer interaction (HCI) ~Interaction techniques ~Auditory feedback •Human-centered computing ~Human computer interaction (HCI) ~HCI design and evaluation methods ~User studies •Human-centered computing ~Ubiquitous and mobile computing ~Ubiquitous and mobile devices ~Smartphones •Applied computing ~Life and medical

sciences ~Health care information systems •Hardware ~Emerging technologies ~Emerging interfaces

Keywords

Tangible interface; Stress relief; Haptic feedback; Auditory feedback; Ubiquitous computing; Mobile health; Affective design; Digital mental health; Emotional design

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1 Introduction

Stress and anxiety are increasingly prevalent in modern life, with global statistics suggesting that nearly half the population will experience a mental health condition during their lifetime [1, 2]. Despite growing awareness, accessible and engaging tools for real-time emotional regulation remain scarce. While mobile apps for mindfulness or cognitive therapy are widely used, many rely heavily on visual attention and lack tactile or embodied interaction [3]. This motivates the development of accessible, comforting tools to support emotional well-being.

Mello addresses this gap by offering a physical, auditorily expressive object leveraging multisensory feedback to support self-regulation through soothing tactile, visual, and auditory stimuli. The design is inspired by findings from sensory therapy, deep pressure stimulation, and auditory calming techniques [4, 5]. It translates these therapeutic principles into an interactive plush toy that supports stress relief in an intuitive, comforting format. The project combines hardware prototyping, mobile application development, and user-centered design to deliver a novel interface for affective self-care.

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2 Related Work

Existing research highlights the effectiveness of sensory engagement in stress reduction. How tactile, auditory, and vibrotactile stimuli contribute to emotional regulation. This section identifies gaps in current interactive products and provides a theoretical foundation for the design and development of Mello, an interactive plush toy that supports emotional well-being through multisensory feedback.

2.1 Soft Fabrics and Materials

Tactile stimulation is widely recognized as a foundational element in stress relief. Soft, cozy fabrics activate the skin’s mechanoreceptors specialized tactile sensors that send calming signals to the brain [6, 7]. This contact has been associated with reduced cortisol levels and activation of the parasympathetic nervous system, contributing to a relaxation response [8]. The skin, as the body’s largest sensory organ, plays a central role in this affective feedback loop.

Materials such as plush textiles and weighted fabrics are central to therapeutic products like sensory blankets and soft toys. These are often employed in autism therapy, trauma recovery, and general emotional regulation to create a sense of safety and comfort through deep touch [9].

2.2 Sound and Auditory Stimuli

Auditory input, particularly nature sounds, and calming music has a significant impact on stress reduction. Sound-based interventions are shown to decrease cortisol levels and slow physiological processes such as heart rate and respiration [10]. According to research in affective computing, natural soundscapes (e.g., rainfall, birdsong, streams) reliably activate the parasympathetic nervous system and modulate stress-related brain regions [11].

Neuropsychological studies have also found that rhythmic auditory stimulation through music can produce effects comparable to certain anxiolytics by altering brainwave activity and enhancing emotional regulation [12]. These findings underscore the utility of sound as a non-invasive, low-effort method for stress relief, particularly when paired with other sensory modalities.

2.3 Vibrations and Haptic Coordination

Vibrotactile feedback is an emerging modality in mental health technologies. Research in HCI and wearable computing suggests that rhythmic vibrations can facilitate emotional grounding and decrease anxiety [13]. Devices such as vibrotactile belts, handheld stress balls, or wearables that deliver patterned haptic pulses have shown promise in modulating arousal states and enhancing mood regulation [14, 15].

The effectiveness of such interventions is often linked to consistency and synchrony in haptic output; steady, rhythmic vibration tends to yield the most calming effect, especially when paired with tactile or auditory cues [16, 17]. This insight is especially relevant for interactive objects like Mello, where layered haptic and auditory feedback are combined.

2.4 Plush Toys and Deep Pressure Stimulation

Plush toys are increasingly recognized as legitimate tools for emotional regulation, particularly when designed to deliver deep pressure stimulation (DPS). DPS is a therapeutic technique that activates the parasympathetic nervous system and promotes the release of oxytocin and serotonin; neurotransmitters linked to feelings of safety and well-being [18].

Cuddling soft plush toys mimics the effects of a hug, offering psychological comfort through embodied interaction. In trauma-informed care, plush toys are used not only with children but also with adults managing anxiety, PTSD, or sensory processing disorders. Their success lies in combining emotional symbolism with biological mechanisms like pressure-based relaxation and neurotransmitter regulation [19, 20].

3 System Overview

While many sensory therapy products, such as weighted blankets, white noise machines, and vibration pads successfully target individual sensory channels, few offer multimodal sensory integration in a single, embodied, interactive form. Furthermore, most lack real-time user control or customizability through mobile application interactions.

Mello addresses this gap by combining tactile softness, rhythmic vibrations, ambient sound, and responsive lighting into an integrated therapeutic object. With mobile app connectivity for personalization and control, Mello represents a new direction in affective computing and sensory therapy, particularly for everyday stress management. Mello consists of two primary components: the plush toy interface and a mobile application.

3.1 Hardware and Physical Design

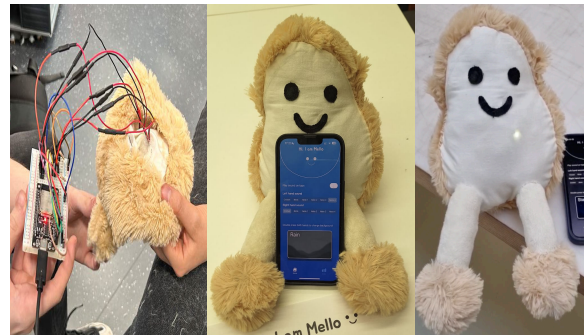


Figure 1: (1) Mello prototype’s inner circuit & wirings. (2) Mello connected with the smartphone application. (3) Prototype affective design - heart glow.

Mello, derived from the word “mellow”, meaning relax, calm, soft, etc. It is a plush toy with an accompanying mobile app. Its size is neither too big, nor too small. The plush is constructed with two types of fabric to indicate the user where to interact with it: a soft cotton front and a furry synthetic back, providing contrasting tactile sensations. Inside the paws, pressure sensors and vibromotors are embedded, allowing for variable haptic feedback. A central pulsing LED simulates slow breathing rhythms, which users can match with their own breath, encouraging paced respiration.

3.2 Embedded Electronics

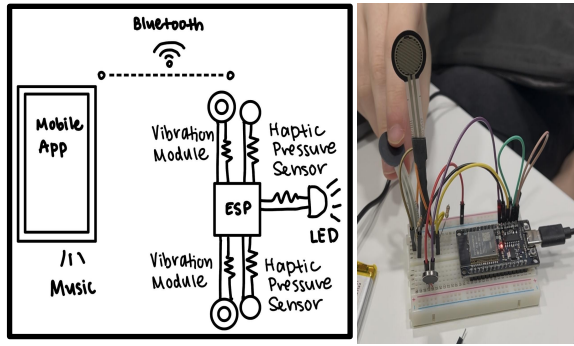


Figure 2: (1) Mello prototype's high-level circuit diagram. (2) Mello embedded connection with microcontroller ESP32, pressure sensors and vibromotors.

An ESP32 microcontroller manages the inputs and outputs, interpreting pressure levels from each paw and modulating motor intensity accordingly. A double-squeeze gesture is mapped to switching ambient background music via the app. The microcontroller communicates via Bluetooth with the smartphone app for synchronization.

3.3 Smartphone Application

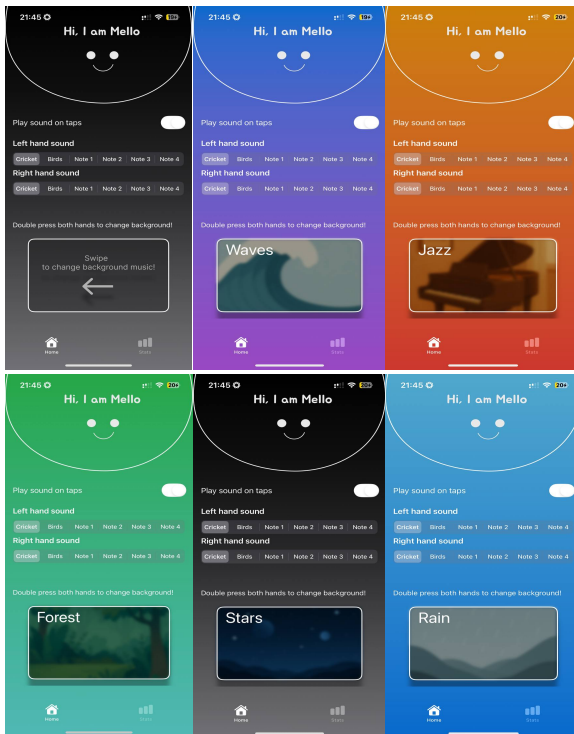


Figure 3: Mello's smartphone application user interface with multiple themes to suit various users and music moods.

Firstly, the app allows the user to switch the sound made by each paw when squeezed. The sound options range from nature sounds like bird chirp or crickets, to different piano notes tuned to the background music. The sound for each paw can be different depending on the user's preference.

Secondly, they can choose different background music to be played. The options include forest noise, waterfall, animal noises and more. Each background music comes with its own appropriate theme, changing the background colour of the app accordingly. Changing the background noise and theme can also be done via squeezing the paws, as will be explained next section 3.4.

Furthermore, the app promotes long-term emotional awareness by showing trends in how and when users interact with Mello, tracking the usage statistics (hourly, daily, weekly). This reinforces mindful habits among users

3.4 Interactions

The first interaction is physically stroking the fabrics to create a soothing sensation and a relaxing experience.

The second interaction is through the "paws" of the toy. There are vibration modules and pressure sensors present in the paws. When the user squeezes the paws, the stronger they squeeze it, the stronger the vibrations produced. This directly correlating experience leads to greater control regarding the strength of the vibrations [21], along with creating an interaction and haptic feedback to keep them grounded and focused on the toy instead of their stress and anxiety.

The third interaction is also through squeezing Mello's paws. When squeezed, each paw makes a sound that can be chosen through the app, as mentioned in Section 3.3. This means that each time any paw is squeezed, it emits a sound. Using this feature, the user can squeeze the paws repeatedly in different rhythms to create music using their chosen sound. If there are 2 users, they can each use 1 paw to create music together.

The fourth interaction is changing the background music coming from the app through squeezing the paws, as mentioned in Section 3.3. When the user squeezes both paws twice, the background music and theme will change on a rotating slider, allowing the user to make the change without needing to slide through and choose on the mobile app.

Another feature of Mello is its light, which pulses in a slow, rhythmic pattern designed to replicate a heartbeat or breathing cycle. This allows users to naturally sync their breathing with the pulsing light, helping them calm down and reduce anxiety or stress. In addition, the combination of ambient nature sounds, and the breathing-pattern light makes Mello an effective night light, supporting relaxation and improving sleep quality. Studies have shown that exposure to calming light rhythms and natural sounds before bedtime can enhance sleep onset and overall sleep quality [22, 23]

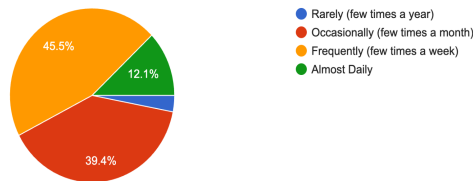
4 Method

A user-centered design approach was utilized to conduct a two-part user evaluation process.

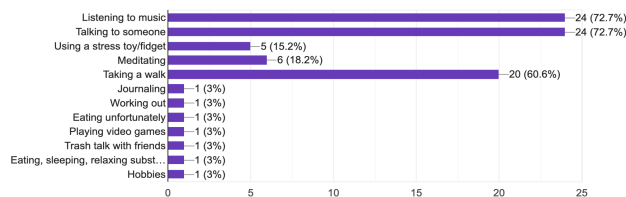
4.1 Pre-Product Survey

An iterative prototyping informed the plush's form, texture, and interaction design

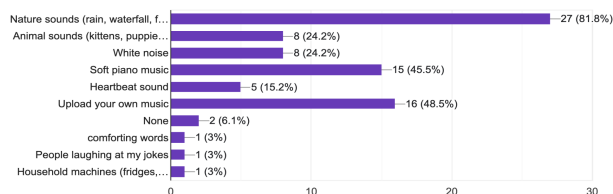
How often do you feel stressed?
33 responses



When you're stressed, what usually helps you feel better? (Select all that apply)
33 responses



What types of sounds help you relax? (Select all that apply)
33 responses



What features would you find most comforting or helpful in such a toy?
33 responses

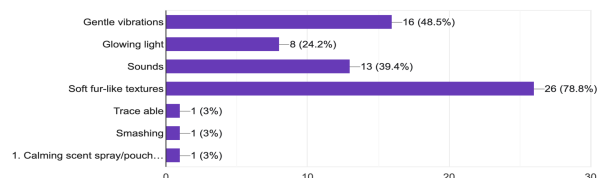


Figure 4: (1) The pie-chart shows that more than 45.5% of the user survey responses indicated frequent stress. (2) Users' choices to relieve stress, this helped us identify which modalities to choose for incorporating in Mello. (3) Selection of sounds that users prefer to relieve stress. (4) Selection of features to add to the design of Mello

An anonymous pre-product survey (n=33) was conducted to understand common stress relief behaviors and feature preferences in an unbiased manner. Results showed high usage of music and tactile tools, with 81.8% of users preferring nature sounds and 78.8% rating soft textures as crucial for comfort. The main goal of the survey was to assess how prevalent the issue of stress is, and what aspects help the user to improve it, so we could incorporate this into the design. 32 respondents claimed that they felt stress at least occasionally, with 4 people saying they felt stressed almost daily. This shows that the problem of stress and anxiety is a prevalent issue in society. The most common ways that worked for users to relieve stress were "Listening to Music", "Talking to someone", and "Taking a

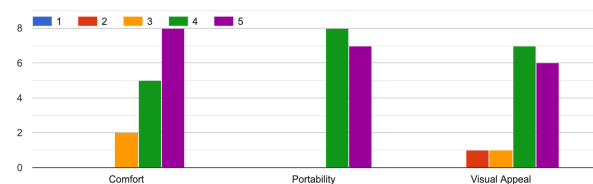
walk". Some also found comfort in using a stress/fidget toy. Additionally, out of 14 respondents who had used plush toys to relieve stress previously, half reported a level 4 of 5 of effectiveness in their use. This proved to us that plush toys and comfort toys would be a beneficial product in reducing stress and anxiety.

The features that people find most helpful in such toys are soft fur-like textures (78.8%), gentle vibrations (48.8%), sounds (39.4%) and glowing light (24.2%). Features such as scent were not considered relevant by the users, which was taken into account for design. The respondents voted in a majority (57.6%) for the product to be medium-sized for hugging.

A very large majority (81.8%) wanted nature sounds like rain, waterfalls, rivers, etc. The next two most voted options were to upload your own music (48.5%) and Soft Piano music (45.5%). This helped us hone in on which type of music to prioritize in our product.

4.2 Post-Product Survey & User Testing

Rate it : 1 (Low) to 5 (High)



Best Features?
15 responses

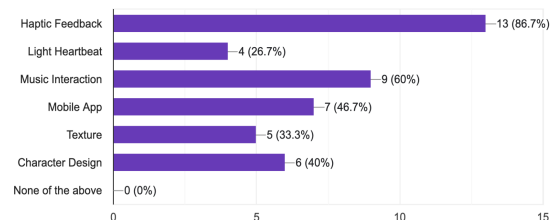


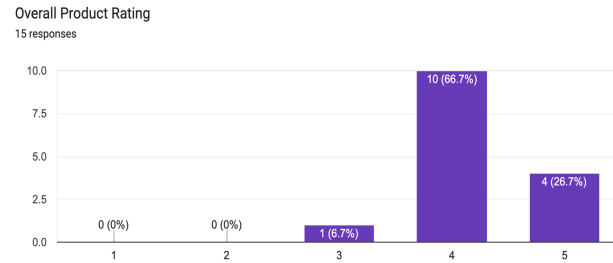
Figure 5: (1) User rating for Comfort, Portability, and Visual Appeal of Mello. (2) Users' selections regarding the best features in Mello.

To get feedback on our final product, we created an online anonymous post-survey feedback form that we enabled on the product demo day. Participants tested the final Mello prototype during a live demo (n=15). They interacted freely, explored features, and completed a post-test survey with both Likert-scale and open-ended questions. We also spoke to users directly about what they thought about the product.

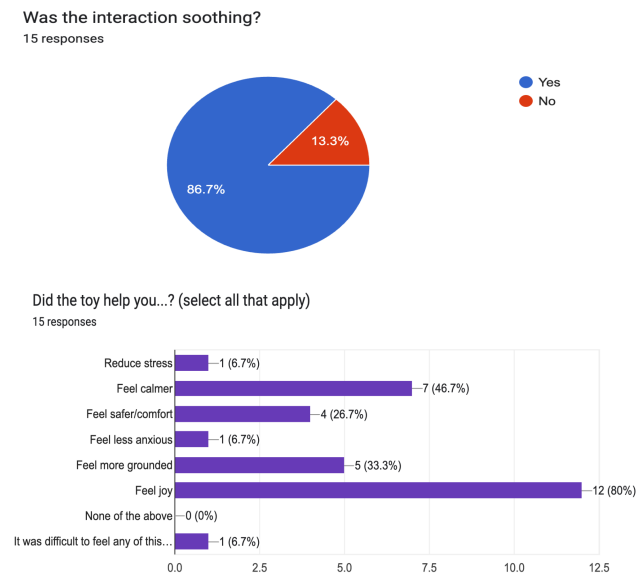
Overall, the response to the product was positive. All key design characteristics of the toy such as perceived comfort, portability, emotional impact, and design appeal received high ratings, with users finding that the best features of the product are "Haptic Feedback", followed by "Music Interaction", and the "Mobile App". The haptic feedback and vibrations emerged as the most-well-received feature, highlighting the effectiveness of the tactile interaction in providing comfort to the users. The lower-voted features like "Light Heartbeat" and "Texture" highlight areas for improvement for us to further enhance the product.

5 Results

An overall product rating of **4.2 out of 5** was achieved.



Emotional impact:



**Figure 6: (1) Overall Product Rating out of 5.
(2) Emotional impact on the User's well-being.
(3) Selection of User's emotional connect with Mello.**

In general, the interaction with the product was considered soothing. Most users felt connected to Mello: 12 felt joy, 7 reported feeling calmer, 5 felt grounded, this shows that the concept hold ground and validity.

Our discussion also revealed that the testing environment also factors in accessing stress, as some participants stated that they weren't feeling very stressed initially, so there wasn't much stress to reduce. Further testing would be necessary to understand the role the initial stress levels play and how to take them into account.

For now, qualitative feedback indicated that the plush was engaging and emotionally supportive and the product has high potential for stress and anxiety relief when used in better and more accurate testing conditions.

6 Discussion & Future Work

Mello shows promise as an accessible, emotionally engaging stress-relief interface. Feedback confirmed the importance of multimodal sensory interaction for emotional support. The combination of textures, sounds, and vibrations allowed users to

engage with Mello in a way that was emotionally satisfying and grounded.

The ability to control Mello without opening the app was also noted as a strength.

Participants requested more light-based customization (color and rhythm), better sound volume control, and broader sound libraries, multi-user remote interaction modes, heat and scent modules for full sensory immersion.

Longitudinal studies will assess effects on stress biomarkers and sleep quality.

7 Conclusion

The design, development, and user testing of Mello - Calm Companion underscore its potential as a viable intervention for stress mitigation and emotional regulation. As a multi-sensory device, it integrates established therapeutic modalities including tactile interaction, rhythmic light patterns, ambient soundscapes, and haptic feedback into a singular, accessible emotional interface.

Insights gathered from both formative and summative user evaluations contributed significantly to refining the product's design and functionality. Participants reported a high level of emotional engagement and perceived soothing effects, indicating the effectiveness of the multi-modal interaction strategy. Moreover, the evaluations revealed key considerations for enhancing usability and personal relevance, particularly concerning sensory preferences and interaction timing.

Furthermore, future work can be done by expanding customization features, improving sensory calibration, and exploring long-term impacts on user well-being. As mental health challenges continue to rise globally, products like Mello represent a promising direction for the integration of design, technology, and psychology in everyday therapeutic tools.

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